



YONGNONG CHEM.

Material Safety Data Sheet

Glufosinate-Ammonium 150g/L SL

1. IDENTIFICATION OF THE SUBSTANCE

Product name: Glufosinate-ammonium 150g/L SL

Company Identification:

Zhejiang Yongnong Chem. Ind. Co., Ltd., Add: Lantian Yongqiang Wenzhou China

Tel: 86-577- 86875158, **Fax:** 86-577-86379678, **P.C.:** 325024

2. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical characterization: phosphinic acid herbicide

Ingredient Name	CAS Number	Exposure Limits	Concentration(g/L)
Glufosinate-ammonium	77182-82-2	OSHA-PEL or ACGIH-TLV: Not established	150
Inert Ingredients			To 1 liter.

3. HAZARD IDENTIFICATION

WARNING: The product is a blue to bluish-green liquid with a weakly pungent odor. Do not get in eyes, on skin, or on clothing. Avoid breathing vapor or spray mist.

Emergency overview: May be fatal if absorbed through skin. Harmful if swallowed or inhaled. A moderate eye irritant. A mild skin irritant. Causes substantial but temporary eye injury.

Primary route(s) of Entry: Skin absorption and spray mist inhalation.

Eye: A moderate eye irritant. Direct contact causes substantial but temporary eye irritation.

Skin: A mild skin irritant. Harmful if absorbed through skin.

Ingestion: Harmful if swallowed.

Effects of overexposure: No incidents of human overexposure are known. Central nervous system stimulation leading to tremors and convulsions occurs in lethally dosed rats.

Signs and symptoms: gastrointestinal disturbance, tremors, convulsions, respiratory depression, cardiac arrhythmia, decreased blood pressure, drowsiness and/or loss of consciousness. These symptoms may be delayed by up to 48h after exposure.

4. FIRST AID MEASURES

General information

Eyes: Flush with plenty of water. Get medical attention if irritation persists.

Skin: Remove contaminated clothing. Wash skin immediately with plenty of soap and water. Get medical attention if irritation occurs.

Ingestion: Rinse mouth thoroughly with plenty of water. Do not induce vomiting. Get medical attention immediately.

Inhalation: Remove individual to fresh air. Get medical attention if breathing difficulty develops.

Advice to doctor

Special notes to a physician: Endotracheal intubation and gastric lavage should be performed as soon as possible, followed by charcoal and sodium sulfate administration. Respiratory, Cardiac and Central Nervous Systems should be monitored with particular regard to ECG, electrolyte balance (especially for potassium) and signs of increased intracranial pressure. In the event of a large exposure, dialysis and/or hemoperfusion should be conducted as soon as possible to eliminate the compound from the body. In the event of convulsions, administer Phenobarbital or diazepam. There is no specific antidote. Glufosinate- ammonium does not inhibit cholinesterase; thus atropine and 2-PAM are contraindicated. Recovery is normally spontaneous, usually within 48h.

5. FIRE-FIGHTING MEASURES

Flash point: $\geq 149^{\circ}\text{F}/65^{\circ}\text{C}$

Suitable extinguishing media:

Small fires: dry chemical, carbon dioxide.

Large fires: Foam or water. Move containers from fire area if without risk. Cool containers with water from maximum distance.

Special equipment for the protection of firefighters: As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH approved (or equivalent) and full protective gear.

Specific methods of fire-fighting: Keep upwind. Isolate hazard area. Avoid inhalation of smoke and fumes. Use water or foam to reduce fumes. Do not touch spilled material. If possible, move containers from area. Extinguish only if flow can be stopped. Use flooding amounts of water as a fog. Cool containers with flooding amounts of water from as far a distance as possible. Avoid breathing vapors.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions: Use proper protective equipment to minimize personal exposure.

Environmental precautions: Take all necessary action to prevent and to remedy the effects of the spill.

Land spill or leak:

Small spill: absorb with an inert absorbent material such as granular clay, saw dust or pet litter. Sweep up carefully while avoiding the formation of a dust cloud. Place in an approved chemical waste container for disposal. Rinse spill area with small amount of soapy water. Contain and absorb the rinsate with inert absorbents and place into the same disposal container. Area can be washed with water to remove the last trace residue. Do not allow water to contaminate water supplies or sewers.

Large spill: Eliminate all ignition sources. Stop leak if you can do so without coming into contact with spilled material. Dike far ahead of liquid spill for later disposal. All equipment used to clean up spill should be grounded. Prevent entry into waterways, sewers, basements or confined areas. Inform appropriate authorities immediately if contamination occurs.

7. HANDLING AND STORAGE

Handling: Remove clothing immediately if pesticide gets inside. Then wash thoroughly and put on clean clothing. Remove Personal Protective Equipment (PPE) immediately after handling this product. Wash the outside of gloves before removing. As soon as possible, wash thoroughly and change into clean clothing. Discard clothing and other absorbent materials that have been drenched or heavily contaminated with this product's concentrate. Do not reuse them.

Specific safe handling advice: Do not apply this product in a way that will contact workers or other persons, either directly or through drift. Only protected handlers may be in the area during application.

Precautions: May be fatal if absorbed through the skin. Harmful if swallowed or inhaled. A moderate eye irritant. A mild skin irritant. Do not get in eyes, on skin, or on clothing. Avoid breathing vapor or spray mist. Wash hands before eating, drinking, chewing gum, using tobacco, or using the toilet.

Storing procedures: Do not use or store near heat or open flame. Do not contaminate water, food, or feed by storage or disposal. Store in original container. Store product in a secure storage area.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Engineering Controls: Handle only in ventilated area.

Eye/face protection: Goggles or face shield.

Skin protection: Impermeable rubber gloves (such as neoprene or PVC).

Respiratory protection: In case of spill, leak, or emergency, use approved chemical cartridge-type respirator.

Other/general protection: Hat, impermeable pants and shirts, and waterproof boots.

For applicators and handlers (agricultural uses): Long-sleeved shirts and long pants, chemical-resistant gloves, shoes plus socks and protective eyewear.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Blue to bluish-green liquid.

Odor: Weakly pungent.

Basic physical properties

Physical state: Liquid.

pH: 6.0 +/- 2 at a concentration of 5% Aqueous Solution.

Density: 1.10 g/cm³ at 20°C

10. STABILITY AND REACTIVITY

Stability: Stable.

Hazardous polymerization: Will not occur.

11. TOXICOLOGICAL INFORMATION

Acute studies

Oral LD₅₀ (rat): >5000 mg/kg (slightly toxic).

Dermal LD₅₀ (rat): > 2000 mg/kg (slightly toxic).

Inhalation LC₅₀ (rat-4hrs): 3.73 mg/l (no more than slightly toxic)

Skin irritation(rat): slightly irritating.

Eye irritation(rabbit): moderately irritating.

Sensitization(Guinea pig): non-sensitizing.

The following data is based on the active ingredient.

Chronic toxicity: No clear evidence of target organ toxicity was noted in studies with rats, mice or dogs.

Carcinogenicity: None.

Teratogenicity: Glufosinate-ammonium, when administered to pregnant animals, produced no adverse effects on fetuses at doses which were not toxic to the dams.

Mutagenicity: No mutagenic activity was detected in a battery of mutagenicity tests.

Neurotoxicity: Does not inhibit acetylcholinesterase activities. No evidence of delayed neurotoxicity was noted in hens. Neurobehavioral effects (e.g., hypersensitivity, tremors, convulsions) related to stimulation of the central nervous system (CNS) were observed in some studies but only at lethal or near lethal dose levels.

12. ECOLOGICAL INFORMATION

Acute and prolonged toxicity to fish: LC₅₀: 79mg/l (96h) in bluegill sunfish, 42mg/l (96h) in rainbow trout, 80mg/l (96h) in carp, 560-1000mg/l in freshwater fish & invertebrates.

Acute toxicity to aquatic invertebrates: EC₅₀/LC₅₀: >100mg/l (48h) in daphnia, 7.2-125mg/l in marine and estuarine organisms.

Chronic toxicity to aquatic invertebrates: NOEC: 32mg/l in freshwater invertebrates.

Toxicity other non mammal terr. Species: LD₅₀/LC₅₀: acute oral >2000mg/kg in avian, acute dietary >5000ppm in avian, reproduction 400ppm in avian, contact >600 µg/bee in honeybee.

Stability in water: hydrolytic half-life: >300 days.

Photodegradation: photolytic half-life: >300 days.

Stability in soil: soil half-life: 6-23 days in field dissipation studies. Koc: 10-1229 (depending on soil type).

Bioaccumulation: no accumulation.

Environmental precautions: Do not apply directly to water, to areas where surface water is present or to intertidal areas below the mean high water mark. Do not clean equipment or dispose of equipment washwaters in a manner that will contaminate water resources of arable land.

Glufosinate-ammonium and its degradates have those properties normally associated with pesticides that have been detected in groundwater. Use of this product in areas with coarse soils and high water tables may result in groundwater contamination.

Ecological information: Rely herbicide is no more than slightly toxic to freshwater fish and aquatic invertebrates. Based on studies conducted with other formulations of glufosinate-ammonium, it is expected to be relatively none toxic to birds, bees, earthworms and beneficial insects, but may be moderately toxic to some marine and/or estuarine organisms.

Environmental fate: Glufosinate-ammonium is very soluble in water and under laboratory conditions is hydrolytically and photolytically stable. Its mobility under laboratory conditions varies widely, depending upon the soil type. However, the results of numerous terrestrial field dissipation studies indicate that the potential for groundwater contaminate is minimal. This appears to be a result of its rapid degradation by microorganisms in the soil and a tendency to bind to certain soil elements resulting no measurable leaching. Glufosinate-ammonium does not accumulate in fatty tissues of fish or other animals.

13. DISPOSAL CONSIDERATIONS

Waste Disposal: Wastes resulting from the use of this product may be disposed of on site or at an approved waste disposal facility.

Container Disposal: Empty containers should be triple rinsed (or equivalent). Then offer for recycling or reconditioning, or puncture and dispose of in a sanitary landfill, or by incineration, or if allowed by Authorities, by burning. If burned, stay out of smoke.

14. TRANSPORT INFORMATION

Proper shipping name: glufosinate-ammonium, Pesticides, liquid, toxic

TDG hazard class: N/A

UN number: N/A

Packing group: N/A

15. REGULATORY INFORMATION

U.S. FEDERAL REGULATORY INFORMATION

SARA TITLE III:

Section 302 (EHS) ingredients: Not listed.

Section 313 ingredients: Not listed.

SARA TITLE III NOTIFICATIONS AND INFORMATION

SARA TITLE III---HAZARD CLASSES: Acute Health Hazard---"Yes"

Chronic Health Hazard---"No"

Fire Hazard---"No"

Sudden Release of Pressure Hazard---"No"

Reactivity Hazard---"No"

EU REGULATORY INFORMATION

EU Classification: None

European Inventory of Existing Commercial Substances (EINECS)

Glufosinate-ammonium 77182-82-2

16. OTHER INFORMATION

This information is provided in good faith but without express or implied warranty. Buyer assumes all responsibility for safety and use not in accordance with label instruction.

ORIGINAL CONTROLLED